

Solution Requirements

Date	15 March 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Accept 7 soil/climate inputs: N, P, K, temperature, humidity, pH, rainfall.	High
2	Authorization levels	Single public user role; no login required to get a recommendation.	Medium
3	External interfaces	Integrate the trained scikit-learn model (model.pkl) for prediction.	High

		<i>services the system must connect to]</i>	
4	Transactions processing	Return one optimal crop out of 22 supported crops per request. <i>will be handled]</i>	High
5	Reporting	Display the recommended crop clearly on the results page. <i>generate]</i>	Medium
6	Business rules, etc.	If a value is unusual, still return the closest suitable crop. <i>process]</i>	Medium
7	Compliance to laws or regulations	No sensitive data collected; inputs are not stored. <i>(e.g., GDPR, HIPAA)]</i>	Low
8	<i>Other</i>	Interactive crop-clicker mini-game and graceful error handling.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Prediction API responds quickly.	< 2 seconds per request
2	Scalability	Handles many farmers via serverless.	Vercel auto-scales on demand
3	Security & Data Privacy	No personal data stored; HTTPS only.	HTTPS + stateless requests
4	Reliability & Availability	Reliable model availability.	~99.9% (Vercel hosting)
5	Usability & Accessibility	Simple, responsive glassmorphism UI.	Loads in < 1 second
6	Other	Maintainable, modular code (PEP 8).	Model accuracy > 90% (97.27%)
		<i>Portability))</i>	